

PVTRACE · DESIGN DECISIONS REGISTER

PVTrace design decisions register

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DOCUMENT CONTROL

Document ID	PVT-DEC-001	Version	0.4
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/pvtrace		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-03	Amish Chadha	Selected capacitor bank minimum recorded as 6.45 mF, accepted by Amish on 2026-10-03; row added; item 9 of the parts to confirm updated
0.3	2026-10-02	Amish Chadha	Cost USD 192 (USD 27 over the target) with the keypad and status LEDs (BOM line 18) and a spare capacitor; R3 met on paper with the selected bank
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations of open items 1 to 6 on 2026-10-02; all moved to decisions made; capacitor selection added to the items to confirm; keypad cost noted
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work; budget treated as a value-engineering target

PVTrace design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The case model: outside size, flat floor, corner pillar positions and size, lid depth of at least 25 mm inside	Sets the plate size and notches, the floor holes and the room for the display and isolator	PVT-DDR-003 C5, C6
2	The display board's size (86 x 50 mm), hole centres (79 x 43 mm) and screen height above the board	Sets the four lid holes and the 7 mm spacer length	PVT-DDR-003 C4
3	The DC isolator's panel cut-out, flange and screw pattern, and depth below the lid (47 mm or less)	Sets the lid holes and the 2.5 mm clearance above the measurement board	PVT-DDR-003 C8, C9
4	The discharge resistor's free-air (unmounted) rating is at least 7 W; if not, mount it on a small aluminium heat spreader	The resistor now sits on a plastic plate; average dump power is 6.6 W at 100 V and one sweep every 5 s	PVT-DDR-003 Table 2; PVT-CAL-001 section 5
5	The fuse holder's base is 44 x 20 mm or less and its DC rating covers 100 V	It sits between the discharge resistor, a corner pillar and the gland nuts with 0.5 to 2 mm to spare	PVT-DDR-003 C1, C6
6	The isolation barrier parts' working voltage and test rating	The barrier must withstand module voltage with margin; checked by the 500 V insulation test	<code>docs/REVIEW.md</code> , TRL 3 safety concerns
7	The USB-C socket has an M16 thread, IP65 or better with a tethered cap	Sets the left end hole and keeps IP54	PVT-DDR-003 C10
8	The modules to be tested have frames 30 to 40 mm deep with a bottom flange	The pod clip grips the frame between its top jaw and thumb screw	PVT-DDR-003 C11
9	The three load capacitors, measured one by one, give at least 6.45 mF together; if not, select others from a larger batch	R3 at the 20 ms lower bound (decided 2026-10-02)	Decision of 2026-10-02 (open item 4); PVT-CAL-001 section 2

■ Value engineering

Value-engineering target: USD 165 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 192 (USD 27 over the target). Main cost drivers and savings worth trying:

- The largest lines are the display board with its ADC module (USD 21), the four load capacitors, three fitted and one spare (USD 24), the DC isolator (USD 16), the measurement board parts (USD 16), the test leads (USD 16) and the sensor pod (USD 16).
- The decisions of 2026-10-02 added USD 14: the keypad and status LEDs (open item 6, new BOM line 18, USD 8) and a spare load capacitor for selecting the bank (open item 4, USD 6).
- Making the design constructable added USD 14: the chassis plate (USD 3), the USB-C socket (USD 5), the pod ADC module (USD 3), the pod clip and gland (USD 1) and fixings (USD 2).
- Savings worth trying: a fused two-pole DC isolator in one body in place of the separate fuse holder and isolator (about USD 5 to 8, if one fits the lid); test leads made from one 2 m pair of PV cable with crimped MC4 connectors (about USD 4); buying the ESP32 board and ADC module as one bundle (about USD 2).

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items 1 to 8: capacitive load; 100 V, 20 A rating; 6.6 mF; isolation option B; 2.8 in TFT plus phone export; once-calibrated reference cell; ESP32 controller; grade thresholds A 90 %, B 80 %, C 70 %	Amish: "i accept all your recommendations, go with them across all repos."	PVT-DDR-001, PVT-DDR-002
2026-09-25	budget_usd set to 165 (item 9); display sun hood and light grey case (item 12); 220 x 130 x 80 mm case (item 13); pitch and problem unchanged	Amish, same instruction	PVT-DDR-002
2026-09-26	PVTrace chosen for the first batch of product renders	Amish	docs/REVIEW.md , 2026-09-26 session
2026-10-01	budget_usd is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	.kit/STANDARDS.md section 18; this register
2026-10-01	Design changed so it can be built (PVT-DDR-003 C1 to C11), under Amish's 2026-09-30 instruction: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." Accepted on 2026-10-02 (below)	Made under Amish's instruction	PVT-DDR-003
2026-10-02	Open item 1: design for construction accepted: C1 to C11 and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	PVT-DDR-003, Tables 1 and 2
2026-10-02	Open item 2: the sensor pod is read on the controller side of the isolation barrier through its own 16-bit ADC (about USD 3, as modelled)	Amish: "i approve your recommendations for all 555 open decisions."	PVT-DDR-003, A1
2026-10-02	Open item 3: the R12 mass margin of about 14 g is accepted, and the prototype is weighed at TRL 4	Amish: "i approve your recommendations for all 555 open decisions."	PVT-DDR-003, A2
2026-10-02	Open item 4: R3 at worst-case tolerance: each capacitor is measured at build and a set is selected that gives at least 6.4 mF (restated as 6.45 mF on 2026-10-03, below); the 20 ms bound is relaxed later only if TOPCon and HJT capacitance data justify it	Amish: "i approve your recommendations for all 555 open decisions."	PVT-DDR-001 item 11; PVT-DDR-002 Table 2
2026-10-02			

Date	Decision	Decided by	Record
	Open item 5: first co-design partner, the first candidate type to approach (not yet agreed): a second-life panel refurbisher or recycler that tests used modules in volume, with a TVET solar course as a good second partner for the teaching use	Amish: "i approve your recommendations for all 555 open decisions."	PVT-DDR-001 item 10; PVT-DDR-002 Table 2
2026-10-02	Open item 6: the case shape items of the appearance model are accepted (corner radii, corner boots, side grips, USB-C flap, labels, lead routing, pod fillets); the keypad and status LEDs are kept, and a BOM line is added for them	Amish: "i approve your recommendations for all 555 open decisions."	docs/REVIEW.md , 2026-09-26 session, items 1 to 4
2026-10-03	Selected load capacitor bank of 6.45 mF or more (the 6.4 mF minimum of 2026-10-02 gives 19.9 ms on the high-current case against the 20 ms bound; 6.45 mF gives 20.05 ms), as recommended	Amish: "Tight Margins - i accept the margins"	PVT-DEC-001 item 4 of 2026-10-02; PVT- CAL-001 section 2; REVIEW.md , session 2026-10-03